INDUSTRIAL INTERNET OF THINGS COURSE END PROJECT

Industrial Internet Of Things(22CIC07N)

Report on

SmartBus Rescue and Access Management System

Submitted by

Dendi Bhavya Reddy – 1601-24-749-005

Kadarla Ramyasri – 1601-24-749-009

M. Deekshitha – 1601-24-749-012

B. Durga Vara Prasad – 1601-24-749-026

B.S. Satwik – 1601-24-749-029

V Sem, CSE (IoT & Cyber Security including Blockchain Technology)

2026-2027

Under Guidance of

Smt. N Sujata Gupta, Asst. Professor,

Dept of CSE(IOT & CSBCT)

Submitted to

Department of CSE(IOT & CSBCT)

Chaitanya Bharathi Institute of Technology (A)

(Affiliated to Osmania University)

Gandipet, Hyderabad – 500075

2026-2027

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

An Autonomous Institute | Affiliated to Osmania University

Kokapet Village, Gandipet Mandal, Hyderabad, Telangana-500075, www.cbit.ac.in

Approved by | Affiliated to | UGC Autonomous | 12 Programs Accredited by NBA | Grade A++ in NAAC | All India Ranking 151-200 Band nirf | Certified by ISO 21001 : 2018

COMMITTED TO RESEARCH, INNOVATION AND EDUCATION 47 years

CERTIFICATE

This is to certify that the **Industrial Internet Of Things(22CIC07N) course end project** titled "**SmartBus Rescue and Access Management System"** is a Bonafide work carried out by **Dendi Bhavya Reddy[160124749005], Kadarla Ramyasri[160124749009], M. Deekshitha[160124749012], B. Durga Vara Prasad[160124749026], B.S. Satwik[160124749029]** students of the Department of CSE(IOT & CSBCT) Chaitanya Bharathi Institute of Technology, Hyderabad, Telangana (India) submitted under the partial fulfilment of the requirements for the award of the degree in Bachelor of Engineering in Academic year 2026 – 2027.

Coordinator

Ms. N. Sujata Gupta

Assistant Professor, Department of CSE(IOT & CSBCT) , CBIT

Place: Hyderabad, Telangana

Date: 31-08-2026

Head of the Department

Dr. Sangeeta Gupta
Professor, CET Dept, CBIT

Table of Contents

S. No.	Content	Page No
1	Title Page	1
2	Certificate	2
3	Table of Contents	3
4	Abstract	4
5	Sensors	4
6	Implications	12
7	Working	15
8	AI Generated Image	21
9	Outcomes	23
10	Drawbacks	25
11	Conclusion and Future scope	26
12	References	27

SmartBus Rescue and Access Management System

1.Abstract

SmartBus Rescue and Access Management System is an IIoT-based smart college transportation system that combines RFID-based student identification, facial verification, bus and route authorization, GPS tracking, and emergency rescue management. The system is designed to provide secure and efficient student transportation while reducing manual verification and improving response during unexpected situations.

During normal boarding, students use an RFID card for identification and a camera for facial verification. An ESP32-S3 controller processes the verification and checks the student's assigned bus and authorized route. The access result is displayed through an OLED, with LEDs and a buzzer providing status indications. All boarding activities can be recorded through IoT-based communication and centralized monitoring.

The system also supports missed-bus and emergency situations. When a student misses the assigned bus, temporary authorization can be provided for a suitable alternate bus without permanently changing the student's assignment. In case of a bus breakdown, the driver or conductor can activate an SOS button. The ESP32 obtains the bus location using a NEO-6M GPS module and sends the information to the backend through Wi-Fi. The backend identifies actually boarded students, checks nearby suitable buses, verifies route compatibility, and calculates available rescue capacity based on current and expected passengers.

Affected students can be allocated across multiple suitable buses when required. Temporary rescue authorization allows students to re-verify using their existing RFID cards and facial verification. Thus, the system provides secure normal access, flexible alternate transportation, and dynamic route-aware emergency rescue management.

2. Sensors and Input Components

Sensors and input components are an essential part of the **SmartBus Rescue and Access Management System**, as they allow the system to identify students, verify their identity, determine bus location, and detect emergency conditions. The system primarily uses an **RFID reader, camera module, GPS module, and SOS button** as input devices. These inputs are processed by the ESP32-based control unit and transmitted through Wi-Fi for centralized monitoring, authorization, and emergency management.

2.1 RFID Reader

The **RC522 RFID reader** is the primary identification component used in the system. It

reads the unique identification number of the student's RFID card and sends the information to the ESP32 controller through the **SPI communication interface**.

Each registered student is provided with an RFID card containing a unique card identity. The corresponding RFID UID is linked to the student's information in the central database, including the assigned bus and authorized route.

When a student approaches the bus and taps the card near the RFID reader, the system reads the card UID and identifies the student.

The RFID reader performs two important functions:

1. **Student Identification** – Identifies the student using the unique RFID card UID.
2. **Boarding Authentication** – Provides the identity information required for bus and route authorization.

The RFID reader therefore acts as the first authentication layer of the system.

2.2 RFID Card

The **MIFARE Classic 1K RFID card** is used as the student's identification card. Each card has a unique UID that is associated with the registered student's information in the database. The card does not need to store the complete student information. Instead, the UID can be used as a key to retrieve the student's details from the backend database.

The information associated with the RFID identity may include:

- Student ID
- Student name
- Assigned bus
- Authorized route
- Authorized stops or areas
- Transportation status

During boarding, the card UID is read by the RC522 reader and passed to the ESP32 for further verification.

2.3 Camera Module for Facial Verification

The **ESP32-S3-WROOM-1 N16R8 CAM** with an **OV2640 camera** is used to capture the student's face for identity verification.

Facial verification provides a second authentication layer after RFID identification. This helps prevent situations where a student uses another student's RFID card.

The verification process can be represented as:

RFID Card → Student Identification → Face Capture → Facial Verification → Identity Match

For example, if the RFID card identifies the student as S101, the captured face must also match the registered facial identity of S101. If the identities match, the system proceeds with bus authorization. If they do not match, access is denied.

The camera therefore improves the security of the student boarding process.

2.4 GPS Module

The **NEO-6M GPS module** is used to obtain the geographical location of the bus. It provides information such as latitude, longitude, and time.

GPS is particularly important during **emergency rescue management**. When a bus experiences a breakdown, its current location can be transmitted to the backend system. The location information can then be used to identify suitable nearby buses for rescue operations.

GPS information can support:

- Bus location tracking
- Emergency bus location
- Identification of nearby buses
- Rescue coordination
- Route comparison
- Transport administrator monitoring

The GPS module does not directly detect a mechanical breakdown. Instead, the **SOS button** is used to report the emergency, while GPS provides the location of the affected bus.

2.5 SOS/Emergency Button

The **SOS button** is used as an emergency input for reporting situations such as bus breakdowns or other transportation emergencies.

When the driver or conductor activates the SOS button, the ESP32 detects the input and obtains the current bus location from the GPS module. The emergency information is then transmitted through Wi-Fi to the backend system.

The emergency sequence can be represented as:

SOS Button Press → ESP32 Detection → GPS Location → Wi-Fi Transmission → Backend → Rescue Management

After receiving the emergency event, the backend can identify the students who have actually boarded the affected bus and begin the rescue allocation process.

2.6 ESP32 as the Control and IoT Unit

The **ESP32-S3-WROOM-1 N16R8 CAM** acts as the main processing and communication unit of the proposed system. It receives information from the RFID reader, camera, GPS module, and emergency input and coordinates the overall operation.

The ESP32 can:

- Receive RFID card information.
- Coordinate facial verification.
- Identify registered students.
- Check assigned bus and route information.
- Process emergency inputs.
- Obtain GPS information.
- Control the OLED display.
- Control LEDs and buzzer.
- Record boarding events.
- Communicate with the backend through Wi-Fi.

The ESP32 is suitable for the project because it combines processing capability with built-in Wi-Fi communication. The ESP32 acts as the central controller for authorization, emergency handling, display control, event recording, and IoT communication.

2.7 OLED Display

The **0.96-inch OLED display** is used to provide visual feedback to students and transport staff during the verification process.

The display can show messages such as:

- **Scan Card**
- **Verifying...**
- **Access Granted**
- **Access Denied**
- **Wrong Bus**
- **Face Mismatch**
- **Emergency Mode**
- **Alternate Bus**
- **Rescue Boarding**

The OLED does not make the authorization decision. Instead, it communicates the current system status to the user and improves the usability of the prototype.

2.8 LED and Buzzer Indicators

LEDs and a buzzer provide local visual and audio indications of the access decision.

A **green LED** can indicate successful authorization, while a **red LED** can indicate denial or an error condition. The buzzer can provide an alert for unauthorized access, invalid RFID cards, face mismatch, or wrong-bus attempts.

For example:

Valid RFID + Valid Face + Authorized Bus → Green LED / Access Granted

Invalid Card / Face Mismatch / Wrong Bus → Red LED + Buzzer / Access Denied

These indicators allow the student and bus in-charge to immediately understand the result of the verification process.

2.9 Sensor and Input Selection Considerations

The selection of sensing and input components has a direct effect on the reliability of the SmartBus system. The following factors should be considered:

Identification Accuracy:

The RFID reader should reliably detect registered cards and obtain their UID.

Face Verification Accuracy:

The camera should capture sufficient facial information for reliable identity verification.

GPS Accuracy:

The GPS module should provide suitable location information for bus tracking and emergency coordination.

Response Time:

The components should respond quickly enough to avoid unnecessary delays during student boarding.

Reliability:

The components should operate reliably under regular bus usage conditions.

Environmental Conditions:

The camera, RFID reader, wiring, and other components may be exposed to dust, vibration, lighting variations, and frequent student interaction.

Electrical Compatibility:

The output and communication interfaces of the components must be compatible with the ESP32 and supporting circuits.

Data Security:

Student identification, facial information, boarding records, and transportation data should be securely handled.

2.10 Overall Sensor and Input System

The sensors and input components work together as a complete transportation monitoring and verification system. During normal boarding, the student first presents the RFID card. The RC522 reader obtains the card UID, after which the camera captures the student's face for verification.

The ESP32 processes these inputs and checks the student's assigned bus and authorized route. If the verification is successful, access is granted and the boarding event can be recorded through the IoT system.

During an emergency, the SOS button provides the emergency input, while the GPS module provides the affected bus location. The information is transmitted through Wi-Fi to the backend, where emergency and rescue operations can be managed.

The overall sensing and input process can be summarized as:

Student
↓
RFID Card
↓
RC522 RFID Reader
↓
Student Identification
↓
Camera / Facial Verification
↓
ESP32 Processing
↓
Bus and Route Authorization
↓
Access Decision
↓
IoT Logging

For emergency situations:

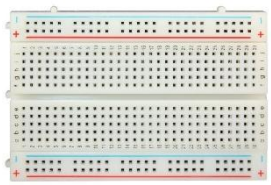
SOS Button
↓
ESP32
↓
NEO-6M GPS
↓
Bus Location
↓
Wi-Fi Communication
↓
Backend
↓
Route-Aware Rescue Allocation

Thus, the sensing and input system provides the connection between the **physical bus environment and the digital IIoT management system**. It enables secure student identification, facial verification, bus tracking, emergency reporting, and centralized transportation monitoring.

Component Table

S.No.	Component	Function	Image
1	**RC522 / MFRC522 RFID Reader**	Reads the unique UID of the student's RFID card and provides the identification information required for student authentication and bus authorization.	
2	**MIFARE Classic 1K RFID Card**	Acts as the student's identification card. Its unique UID is linked to the student's information in the central database.	
3	**ESP32-S3-WROOM-1 N16R8 CAM**	Main processing and IoT communication unit. It coordinates RFID, facial verification, GPS, emergency inputs, authorization, display, indicators, and Wi-Fi communication.	
4	**OV2640 Camera**	Captures the student's face for facial identity verification after RFID identification.	
5	**NEO-6M GPS Module**	Provides bus latitude, longitude, and time information for bus tracking and emergency rescue coordination.	

S.No.	Component	Function	Image
6	**0.96-inch OLED SSD1306**	Displays system messages such as Scan Card, Verifying, Access Granted, Access Denied, Wrong Bus, and Emergency Mode.	
7	**SOS / Emergency Button / push button**	Allows the driver or conductor to report a bus breakdown or emergency condition to the ESP32.	
8	**Green LED**	Provides a visual indication of successful authorization or access granted.	 -
9	**Red LED**	Provides a visual indication of access denial, verification failure, or error conditions.	
10	**3.3V Active Buzzer**	Provides an audio alert for unauthorized access, invalid cards, face mismatch, or wrong-bus attempts.	
11	**220 Ω Resistors**	Used with indicator LEDs to limit current and protect the components.	

S.No.	Component	Function	Image
12	**Breadboard and Jumper Wires**	Used for electrical connections and prototype-level integration of the hardware components.	 
13	**USB-C Data Cable**	Used to power the ESP32 development board, upload the program, and provide USB communication during prototype development.	

3. Implications

SmartBus Rescue and Access Management System has applications in secure college transportation, automated student verification, IIoT-based monitoring, emergency response, and intelligent bus management. The project demonstrates how embedded systems, authentication technologies, GPS, wireless communication, and centralized data management can be integrated to improve both normal and emergency transportation operations.

3.1 Secure Student Transportation

The system improves the security of college bus transportation by combining **RFID-based identification and facial verification**. This helps reduce unauthorized card usage, card sharing, and boarding of students onto buses for which they are not authorized. The system can also verify the assigned bus and

permitted route before allowing boarding.

3.2 Automated Bus Access Management

The proposed system reduces the need for continuous manual verification by providing an automated access decision. The ESP32 processes the RFID and facial-verification results and provides **Access Granted** or **Access Denied** indications through the OLED, LEDs, and buzzer.

3.3 IIoT-Based Transportation Monitoring

The system connects physical bus operations with centralized digital monitoring. Boarding information such as **student ID, bus ID, route, time, verification status, and access result** can be transmitted through Wi-Fi and stored for transport management and analysis.

The basic flow is:

Student → RFID + Face Verification → ESP32 → Authorization → Wi-Fi → IoT Monitoring

3.4 GPS-Based Bus Tracking

The integration of GPS provides the location of buses and becomes particularly useful during emergency situations. Bus location information can help transport administrators identify the affected bus and coordinate suitable alternate or rescue buses.

GPS information can support:

- Bus location tracking
- Emergency location identification
- Nearby bus identification
- Rescue coordination
- Route comparison
- Centralized transport monitoring

3.5 Missed-Bus Management

The system can provide flexibility when a student misses their assigned bus. Instead of permanently changing the student's bus assignment, a **temporary**

alternate-bus authorization can be created for a suitable permitted bus. This allows the student's existing RFID identity to remain unchanged while providing temporary access according to the configured transportation rules.

3.6 Emergency Rescue Management

A major implication of the proposed system is its ability to support **bus-breakdown and emergency situations**. When an emergency is reported, the system can identify students who have actually boarded the affected bus and coordinate their transfer to suitable rescue buses.

The backend can consider:

- Affected students
- Bus location
- Route compatibility
- Current passengers
- Pending assigned students
- Already allocated rescue students
- Available operational capacity

If one suitable bus cannot accommodate all affected students, the students can be distributed among multiple suitable buses.

3.7 Dynamic Capacity Management

The system can use available passenger capacity to determine how many affected students can be assigned to a rescue bus.

The available rescue capacity can be represented as:

Rescue Capacity = Maximum Operational Capacity − Current Passengers − Pending Assigned Students − Already Allocated Rescue Students

This allows rescue allocation to consider not only the students requiring rescue but also passengers who are already travelling or still expected to board.

3.8 Temporary Rescue Authorization

The system provides temporary authorization for students using an alternate or rescue bus without modifying their permanent bus assignment.

During rescue boarding, the student can again use:

RFID → Facial Verification → Temporary Authorization → Route Check → Rescue Boarding

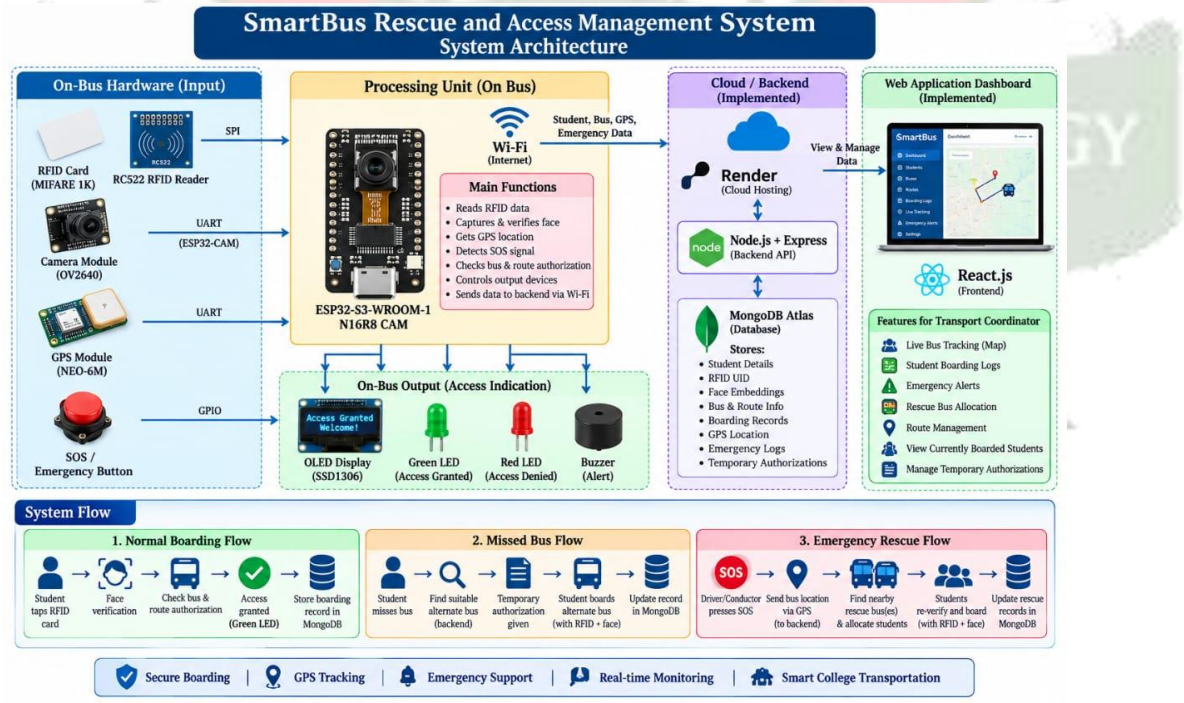
The rescue boarding event can then be recorded to maintain an updated transportation status.

3.9 Potential Applications

The system can be considered for:

- College and university transportation
- School bus management
- Institutional employee transport
- Route-based private transportation
- Emergency bus coordination
- Smart transportation infrastructure
- Centralized fleet monitoring

Overall, the project demonstrates how **RFID, facial verification, GPS, ESP32, wireless communication, and intelligent transportation management** can be combined to create a secure and flexible IIoT-based college bus system. It extends normal access control with **missed-bus handling, temporary alternate authorization, and dynamic route-aware emergency rescue management**.



4.Working

The working of the **SmartBus Rescue and Access Management System** is based on secure student identification, facial verification, bus and route authorization, IoT-based boarding logging, GPS-based bus tracking, and emergency rescue management.

The overall process is:

Student → RFID Identification → Face Verification → Bus/Route Check → Access Decision → Boarding Log

For emergency situations:

SOS → GPS Location → Backend → Affected Students → Suitable Rescue Buses → Temporary Authorization → Rescue Boarding

4.1 Student Identification

When a student approaches the bus, the student taps the registered **RFID card** on the **RC522 RFID reader**.

The RFID reader obtains the unique UID of the card and sends it to the ESP32. The UID is used to identify the corresponding student from the database.

RFID Card → RC522 Reader → RFID UID → Student Identification

The system then proceeds to facial verification.

4.2 Facial Verification

After identifying the student through RFID, the **OV2640 camera** captures the student's face. The system verifies whether the captured face matches the registered facial identity associated with the RFID card.

For example:

RFID → S101

Face → S101

If both identities match, the system continues with bus authorization. If the face does not match, access is denied.

This provides an additional layer of security and helps prevent the use of another student's RFID card.

4.3 Bus and Route Authorization

After successful identity verification, the system checks the student's transportation information.

It verifies whether:

- The student is authorized for the current bus.

- The current bus operates on a permitted route.
- Any applicable temporary authorization exists.

If the student is authorized, the system generates an **Access Granted** decision.

If the student attempts to board an unauthorized bus, the system generates an **Access Denied** decision.

Identity Verified → Bus Check → Route Check → Access Decision

The system uses assigned-bus and authorized-route information for access decisions.

4.4 Access Indication

The ESP32 provides the access result through the connected output devices.

For successful boarding:

Access Granted → OLED Message + Green LED

For an unauthorized attempt:

Access Denied → OLED Message + Red LED + Buzzer

The OLED can display messages such as **"Scan Card," "Verifying...," "Access Granted," "Access Denied," "Wrong Bus," and "Face Mismatch."**

4.5 Normal Boarding and IoT Logging

When access is granted, the student boards the bus.

The boarding event can then be recorded and transmitted through **Wi-Fi** to the backend or database.

The system can record information such as:

- Student ID
- Bus ID
- Route
- Date and time
- Verification status
- Access result
- Boarding type

This creates a digital record of student transportation activity.

4.6 Missed-Bus Handling

The system also supports situations where a student misses the assigned bus.

The student's permanent bus assignment is not changed. Instead, when a suitable alternate bus is available, the backend can create a **temporary alternate-bus authorization**.

For example:

Student S101 → Assigned Bus B1

If S101 misses B1 and B2 is a permitted alternate bus:

S101 → Temporary Authorization → B2

The student can then use the existing RFID card and complete the required verification on the alternate bus.

4.7 Emergency Detection

If an assigned bus experiences a breakdown or emergency, the **driver or conductor can press the SOS button**.

The ESP32 detects the emergency input and obtains the current location of the bus using the **NEO-6M GPS module**.

The emergency information is then transmitted through Wi-Fi to the backend.

SOS Button → ESP32 → GPS Location → Wi-Fi → Backend

The GPS provides the location of the bus; it does not itself determine that a mechanical breakdown has occurred.

4.8 Identification of Affected Students

After receiving the emergency information, the backend identifies the students who have **actually boarded the affected bus** using the boarding records.

For example:

Students Assigned to B1 = 40

Students Actually Boarded B1 = 27

In this situation, the rescue operation is initially concerned with the **27 students who actually boarded the bus**, rather than automatically treating all 40 assigned students as being on the bus.

4.9 Finding Suitable Rescue Buses

The backend uses available bus information and GPS locations to identify suitable buses for rescue.

However, the system does not select a rescue bus based only on distance. It also considers:

- Route compatibility
- Destination or permitted travel area
- Current passengers
- Pending assigned students
- Already allocated rescue students
- Maximum operational capacity

This allows rescue allocation to consider the actual transportation situation of each bus.

4.10 Dynamic Rescue Capacity Calculation

For each suitable rescue bus, the system calculates how many additional students can be accommodated.

Maximum operational capacity refers to the maximum number of passengers that a bus can safely accommodate under its configured operating conditions, including seated capacity and any permitted standing capacity. For example, a bus with 25 seats and permission for 5 standing passengers has a maximum operational capacity of 30 passengers.

The calculation is:

Rescue Capacity = Maximum Operational Capacity − Current Passengers − Pending Assigned Students − Already Allocated Rescue Students

For example, if a bus has an operational capacity of 30, currently has 15 passengers, has 5 pending assigned students, and already has 4 rescue students allocated:

Rescue Capacity = 30 − 15 − 5 − 4 = 6 students

The backend can use this available capacity when allocating affected students.

4.11 Multi-Bus Rescue Allocation

If one suitable bus cannot accommodate all affected students, the backend can distribute the students among multiple suitable buses.

For example:

Affected Students → Rescue Bus B2 + Rescue Bus B3

The allocation is based on available capacity and route or destination compatibility.

This prevents the system from assigning more students to a rescue bus than its configured operational capacity allows.

4.12 Temporary Rescue Authorization

After rescue allocation, the backend creates a temporary authorization for each affected student.

The authorization can contain information such as:

- Student ID
- Original bus
- Temporary rescue bus
- Purpose
- Creation time
- Authorization status

The student's original bus assignment remains unchanged.

At the rescue bus, the student uses the same RFID card and completes facial verification.

RFID → Face Verification → Temporary Authorization → Route Check → Rescue Boarding

Once the rescue boarding is confirmed, the boarding event is recorded.

4.13 Transportation Status Update

The system can maintain the student's transportation status throughout the journey.

For normal transportation:

ASSIGNED → VERIFIED → BOARDED → TRAVELLING

For a missed-bus situation:

ASSIGNED → MISSED BUS → ALTERNATE AUTHORIZATION → ALTERNATE BOARDED

For an emergency situation:

BOARDED → BUS BREAKDOWN → RESCUE ALLOCATED → RESCUE BOARDED

This provides the transport administrator with an updated view of the student's transportation status.

4.14 Fest Mode

Fest Mode is a software-based transportation policy mode used during college festivals or special events when normal bus assignments may need temporary flexibility. In this mode, authorized students can be permitted to use selected alternate buses according to configured route, destination, and capacity rules. The student's permanent bus assignment is not changed. Instead, temporary authorization is created for the duration of the event and all alternate boarding activities are recorded through the IoT system. RFID and facial verification continue to be used to maintain secure student identification.

4.15 Complete Working

The complete normal operation can be summarized as:

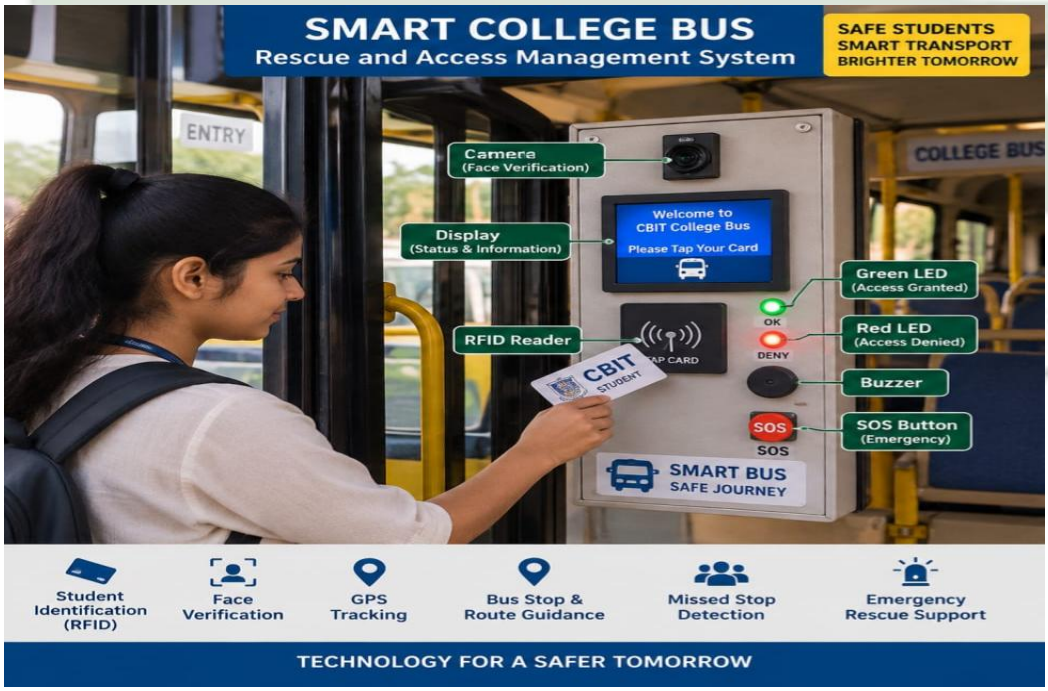
1. Student approaches the bus.
2. Student taps the RFID card.
3. RC522 reads the RFID UID.
4. ESP32 identifies the student.
5. Camera captures and verifies the student's face.
6. System checks the assigned bus and authorized route.
7. ESP32 generates the access decision.
8. OLED, LED, and buzzer provide the status indication.
9. If access is granted, the student boards the bus.
10. Boarding information is transmitted through Wi-Fi and recorded.

The emergency operation can be summarized as:

1. Bus experiences a breakdown or emergency.
2. Driver or conductor presses the SOS button.
3. ESP32 detects the emergency.
4. GPS provides the bus location.
5. Emergency information is sent to the backend.
6. Backend identifies students who actually boarded the affected bus.
7. Suitable rescue buses are identified.
8. Route compatibility and available capacity are checked.
9. Students are allocated to one or more rescue buses.
10. Temporary rescue authorizations are created.
11. Students re-verify using RFID and facial verification.
12. Rescue boarding is recorded.
13. Transportation status is updated.

Thus, the system combines **secure normal boarding with missed-bus handling and dynamic emergency rescue management**. The important feature is that the backend does not simply allow students onto another bus; it determines whether the alternate bus is suitable based on **route compatibility and available operational capacity**, and then records the student's temporary rescue boarding.

5.AI Generated Image



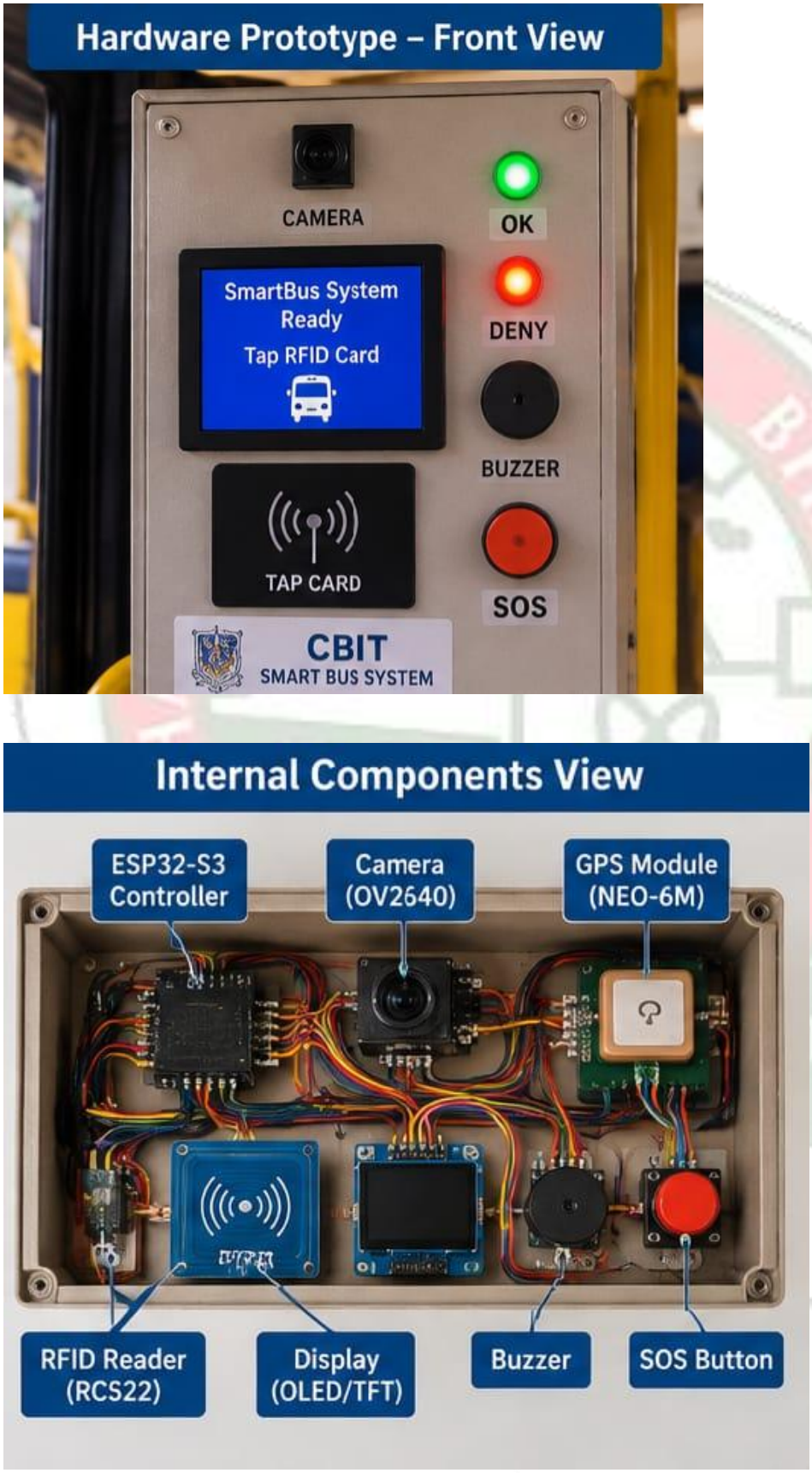


Figure 1: AI-Generated Prototype of the SmartBus Rescue and Access Management System

6.Outcomes

The expected outcome of the **SmartBus Rescue and Access Management System** is a smart college transportation system that can **securely verify students during normal boarding while also managing missed-bus and emergency rescue situations**. The project aims to demonstrate the practical integration of RFID, facial verification, GPS, ESP32-based control, and IIoT-based transportation monitoring.

6.1 Student Identification and Verification

The system is expected to identify students using their registered **RFID cards** and verify their identity using facial verification. Only when the RFID identity and facial identity match can the system proceed with bus authorization.

6.2 Automated Bus and Route Authorization

The system checks whether the verified student is authorized to board the current bus according to the assigned bus and permitted route. A valid combination results in **Access Granted**, while an invalid card, face mismatch, or unauthorized bus results in **Access Denied** with an appropriate alert.

6.3 Missed-Bus Management

The system is expected to support students who miss their assigned bus by providing **temporary authorization for a suitable alternate bus**. The student's permanent bus assignment remains unchanged, while the temporary authorization allows controlled alternate boarding.

6.4 Emergency and Rescue Management

In case of a bus breakdown or emergency, the system can receive an emergency signal through the **SOS button** and obtain the affected bus location using GPS. The backend can identify students who actually boarded the affected bus and coordinate their transfer to suitable rescue buses.

6.5 Dynamic Rescue Allocation

The system can calculate available rescue capacity by considering the **maximum operational capacity, current passengers, pending assigned students, and already allocated rescue students**.

If one bus cannot accommodate all affected students, the system can distribute them among multiple suitable buses according to **route or destination compatibility**.

6.6 IoT Data Logging

The system is expected to transmit transportation information through Wi-Fi to the backend

or database. The recorded information can include:

- Student ID
- Bus ID
- Route
- Date and time
- Verification status
- Boarding type
- Bus location
- Emergency status

This information can support centralized transport monitoring and management.

6.7 Rescue Boarding Verification

After a student is allocated to an alternate or rescue bus, the student can use the existing RFID card and complete facial verification again. The temporary authorization is checked before allowing rescue boarding.

The process is:

RFID → Face Verification → Temporary Authorization → Route Check → Rescue Boarding

The completed rescue boarding event is then recorded to maintain an updated transportation status.

6.8 Practical Demonstration

The completed prototype is expected to demonstrate the normal boarding process, unauthorized access handling, GPS-based bus location, emergency reporting, temporary alternate-bus authorization, and rescue boarding.

The system can demonstrate how a student's transportation status changes from normal boarding to alternate or rescue transportation when required.

6.9 Overall Outcome

The project is expected to demonstrate:

1. **RFID-based student identification.**
2. **Facial identity verification.**
3. **Automated bus and route authorization.**
4. **Missed-bus temporary authorization.**
5. **GPS-based bus location monitoring.**
6. **SOS-based emergency reporting.**
7. **Dynamic route-aware rescue allocation.**

8. **Capacity-aware multi-bus rescue management.**
9. **Temporary rescue boarding verification.**
10. **Wireless transmission and IoT-based data logging.**

Overall, the project demonstrates how **student authentication, intelligent bus authorization, GPS tracking, and emergency rescue management** can be combined into a single IIoT-based college transportation system. The system extends normal access control by providing a structured mechanism for **alternate transportation and dynamic emergency rescue operations**.

7. Drawbacks

Although the **SmartBus Rescue and Access Management System** improves bus security and emergency management, it has some practical limitations.

7.1 Dependence on Internet

The system depends on Wi-Fi/Internet for sending data to the backend and updating the dashboard. Poor connectivity may delay updates.

7.2 GPS Limitations

GPS may not always provide accurate location information, especially in areas with weak signals or obstructions.

7.3 Face 7Verification Issues

Face verification may be affected by poor lighting, camera position, or changes in the student's appearance.

7.4 RFID Limitations

RFID cards can be lost, damaged, or shared. Therefore, RFID is combined with face verification for better security.

7.5 Emergency Detection

The system depends on the driver or conductor pressing the **SOS button** to report a bus breakdown or emergency.

7.6 Capacity Management

Rescue allocation depends on correct information about current passengers, pending students, and available bus capacity.

7.7 Hardware Maintenance

The RFID reader, camera, GPS, ESP32, OLED, and other components require proper installation and regular maintenance.

7.8 Initial Cost

Deploying the system across many buses requires additional hardware, communication, backend, and maintenance costs.

7.9 Privacy and Security

Student identification and facial data must be stored and handled securely to protect student privacy.

7.10 Overall Limitation

The main challenge is maintaining **accurate verification, GPS tracking, capacity information, and reliable communication** while keeping the system practical and affordable.

8.Conclusion And Future Scope

8.1 Conclusion

The SmartBus Rescue and Access Management System combines RFID, face verification, GPS, ESP32, and IoT technology to improve college bus security and transportation management.

The system verifies students before boarding and checks their assigned bus and route. In case of a missed bus or bus breakdown, it can identify suitable alternate/rescue buses based on route compatibility and available capacity. Temporary authorization allows students to board the selected rescue bus securely.

Overall, the project demonstrates how IIoT, authentication, GPS tracking, and dynamic rescue management can be combined to provide safer and more organized college transportation.

8.2 Future Scope

The system can be further improved by:

- Mobile App: Provide students and administrators with real-time bus information.
- Live Tracking: Improve real-time monitoring of buses and routes.
- Advanced Face Recognition: Improve verification accuracy under different conditions.
- Automated Rescue Allocation: Make rescue-bus allocation faster and more efficient.
- Cloud Analytics: Analyze boarding, route, and emergency data.
- Notifications: Send alerts about missed buses, breakdowns, and rescue assignments.
- Scalability: Extend the system to more buses, routes, and campuses.

Overall, the system can be developed into a complete smart transportation management solution for educational institutions.

9. References

1. **Espressif Systems, "ESP32-S3-WROOM-1 and ESP32-S3-WROOM-1U Datasheet," Espressif Systems.**
2. **Espressif Systems, "ESP-IDF Programming Guide for ESP32-S3," Espressif Systems.**
3. **NXP Semiconductors, "MFRC522 Standard Performance MIFARE and NTAG Frontend Datasheet," NXP Semiconductors.**
4. **u-blox, "NEO-6 u-blox 6 GPS Modules Data Sheet," u-blox.**
5. **Solomon Systech, "SSD1306 128 × 64 Dot Matrix OLED/PLED Segment/Common Driver with Controller," Solomon Systech.**
6. **Espressif Systems, "ESP32-S3 Technical Reference Manual," Espressif Systems.**
7. **Y. Wang et al., "Data-Driven Intelligent Transportation Systems: A Survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 12, no. 4, pp. 1624–1639, 2011.**
8. **"Internet of Things (IoT): A Vision, Architectural Elements, and Future Directions," *Future Generation Computer Systems*.**
9. **"A Survey on Internet of Things Solutions for Intelligent Transportation Systems," *IEEE Access*.**
10. **"Internet of Things for Smart Transportation Systems: A Review," *Sensors*.**